

CUSTOMER SHOPPING BEHAVIOUR ANALYSIS

Project Report

Author: Adithya V S

GitHub: [Adithyasajeev22/Customer-Shopping-Behaviour-Analysis](https://github.com/Adithyasajeev22/Customer-Shopping-Behaviour-Analysis)

Project Type	End-to-End Data Analytics Project
Domain	Retail / E-commerce Customer Analytics
Tools Used	Python, Pandas, NumPy, Matplotlib, Seaborn, MySQL, Excel and Power BI
Project Goal	Analyze customer shopping behaviour and transform the data into meaningful business insights.

1. Executive Summary

Customer Shopping Behaviour Analysis is an end-to-end data analytics project focused on understanding customer purchasing patterns, spending behaviour, product performance, customer satisfaction, subscription status, and demographic characteristics. Python is used for data cleaning and exploratory data analysis, MySQL for business-oriented SQL analysis, and Power BI for interactive visualization.

2. Project Objectives

- Clean and prepare the customer shopping dataset.
- Perform exploratory data analysis.
- Analyze sales and revenue across customer and product dimensions.
- Evaluate customer reviews and satisfaction.
- Analyze discounts, subscriptions, payment methods, and shipping types.
- Answer practical business questions using MySQL.
- Present results through an interactive Power BI dashboard.

3. Dataset Description

The dataset contains demographic, product, transaction, review, subscription, payment, shipping, and purchasing-frequency attributes.

Column	Description
Customer ID	Unique customer identifier
Age	Customer age
Gender	Customer gender
Item Purchased	Product purchased
Category	Product category
Purchase Amount (USD)	Amount spent
Location	Customer location
Size	Purchased product size
Color	Product colour
Season	Purchase season
Review Rating	Customer rating
Subscription Status	Subscription status
Shipping Type	Shipping method
Discount Applied	Whether a discount was applied
Promo Code Used	Whether a promo code was used
Previous Purchases	Number of previous purchases
Payment Method	Payment method
Frequency of Purchases	Customer purchase frequency

4. Project Methodology

The raw dataset was inspected and cleaned in Python. Missing values, duplicates, data types, and inconsistent values were checked. Exploratory analysis was performed using Pandas, NumPy, Matplotlib, and Seaborn. The cleaned dataset was then analyzed using MySQL, and the results were presented using Power BI.

5. Power BI Dashboard

The dashboard contains KPIs for customer count, average review rating, and average purchase amount, together with revenue, sales, subscription, and age-group visualizations.

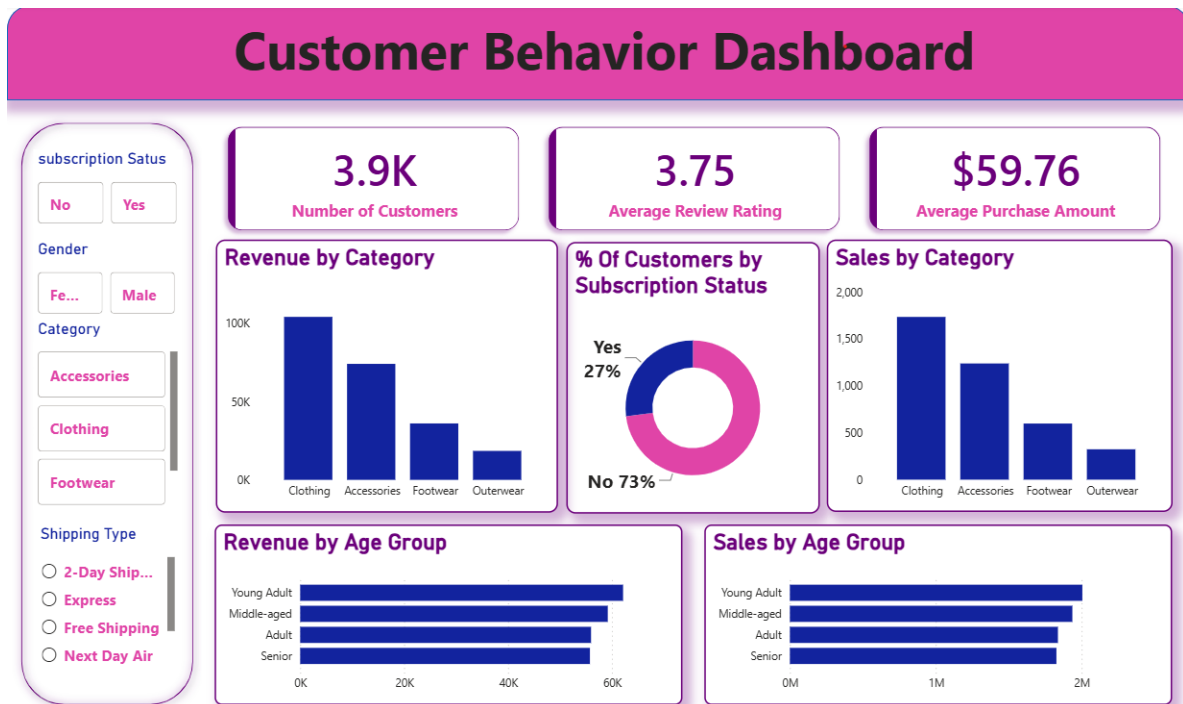


Figure 1. Customer Behavior Dashboard — Power BI

6. MvSQL Business Analysis

SQL Query Analysis and Results

The following SQL queries were developed to answer key business questions from the cleaned customer shopping dataset. Each query is followed by its corresponding MySQL output screenshot, showing the calculated result used in the project's analysis.

```
SELECT gender, SUM(purchase_amount) AS revenue
FROM cleaned_data
GROUP BY gender;
```

SQL Output Screenshot

	gender	revenue
▶	Male	157890
	Female	75191

Result: Male: \$157,890 | Female: \$75,191

Q2. Which customers used a discount but still spent more than the average purchase amount?

```
SELECT customer_id, purchase_amount
FROM cleaned_data
WHERE discount_applied = 'Yes'
AND purchase_amount >= (
  SELECT AVG(purchase_amount)
  FROM cleaned_data
);
```

SQL Output Screenshot

	customer_id	purchase_amount
▶	2	64
	3	73
	4	90
	7	85
	9	97
	12	68
	13	72
	16	81
	20	90
	22	62
	24	88
	29	94
	32	79
	33	67
	35	91
	37	69
	40	60
	44	76

Result: The screenshot lists the customer IDs and purchase amounts satisfying the conditions.

Q3. Which are the top 5 products with the highest average review rating?

```
SELECT
item_purchased,
ROUND(AVG(review_rating), 2) AS average_product_rating
FROM cleaned_data
GROUP BY item_purchased
ORDER BY average_product_rating DESC
LIMIT 5;
```

SQL Output Screenshot

	item_purchased	average_product_rating
▶	Gloves	3.86
	Sandals	3.84
	Boots	3.82
	Hat	3.8
	Handbag	3.78

Result: Gloves 3.86 | Sandals 3.84 | Boots 3.82 | Hat 3.80 | Handbag 3.78

Q4. What is the revenue contribution of each age group?

```
SELECT
age_group,
SUM(purchase_amount) AS total_revenue
FROM cleaned_data
GROUP BY age_group
ORDER BY total_revenue DESC;
```

SQL Output Screenshot

	age_group	total_revenue
▶	Young Adult	62143
	Middle-aged	59197
	Adult	55978
	Senior	55763

Result: Young Adult \$62,143 | Middle-aged \$59,197 | Adult \$55,978 | Senior \$55,763

Q5. Which 5 products have the highest percentage of purchases with discounts applied?

```
SELECT item_purchased,  
ROUND(  
100.0 * SUM(  
CASE WHEN discount_applied = 'Yes' THEN 1 ELSE 0 END  
) / COUNT(*), 2  
) AS discount_rate  
FROM cleaned_data  
GROUP BY item_purchased  
ORDER BY discount_rate DESC  
LIMIT 5;
```

SQL Output Screenshot

	item_purchased	discount_rate
▶	Hat	50.00
	Sneakers	49.66
	Coat	49.07
	Sweater	48.17
	Pants	47.37

Result: Hat 50.00% | Sneakers 49.66% | Coat 49.07% | Sweater 48.17% | Pants 47.37%

Q6. Compare the average Purchase Amounts between Standard and Express Shipping.

```
SELECT shipping_type,  
ROUND(AVG(purchase_amount), 2) AS average_purchase_amount  
FROM cleaned_data  
WHERE shipping_type IN ('Standard', 'Express')  
GROUP BY shipping_type;
```

SQL Output Screenshot

	shipping_type	ROUND(AVG(purchase_amount),2)
▶	Express	60.48
	Standard	58.46

Result: Express \$60.48 | Standard \$58.46

Q7. Do subscribed customers spend more? Compare average spend and total revenue between subscribers and non-subscribers.

```
SELECT subscription_status,  
COUNT(customer_id) AS total_customers,  
ROUND(AVG(purchase_amount), 2) AS avg_spend,  
ROUND(SUM(purchase_amount), 2) AS total_revenue  
FROM cleaned_data  
GROUP BY subscription_status  
ORDER BY total_revenue, avg_spend DESC;
```

SQL Output Screenshot

	subscription_status	total_customers	avg_spend	total_revenue
▶	Yes	1053	59.49	62645
	No	2847	59.87	170436

Result: Yes: 1,053 customers | Avg spend \$59.49 | Revenue \$62,645. No: 2,847 customers | Avg spend \$59.87 | Revenue \$170,436.

Q8. Are customers who are repeat buyers (more than 5 previous purchases) also likely to subscribe?

```
SELECT subscription_status,  
COUNT(customer_id) AS repeat_buyers  
FROM cleaned_data  
WHERE previous_purchases > 5  
GROUP BY subscription_status;
```

SQL Output Screenshot

	subscription_status	repeat_buyers
▶	Yes	958
	No	2518

Result: Yes: 958 repeat buyers | No: 2,518 repeat buyers

7. Key Business Insights

- Male customers generated \$157,890 in revenue compared with \$75,191 from female customers.
- Gloves recorded the highest average review rating among the top five products at 3.86.
- Young Adult customers generated the highest age-group revenue at \$62,143.
- Hat had the highest percentage of discounted purchases at 50.00%.
- Express shipping had a higher average purchase amount (\$60.48) than Standard shipping (\$58.46).
- Subscribers had an average spend of \$59.49, while non-subscribers had an average spend of \$59.87.
- There were 958 repeat buyers among subscribers and 2,518 among non-subscribers.

8. Business Recommendations

- Focus sales and marketing analysis on high-revenue customer segments.
- Monitor highly rated products and maintain their availability and quality.
- Use age-group analysis for targeted customer campaigns.
- Evaluate discount effectiveness at product level.
- Compare shipping options alongside purchase value.
- Strengthen subscription programs to increase subscriber adoption.
- Use repeat-buyer analysis to identify valuable customer segments.

9. Conclusion

The Customer Shopping Behaviour Analysis project demonstrates a complete data analytics workflow using Python, MySQL, and Power BI. The SQL analysis provides measurable answers to business questions involving revenue, customer spending, product ratings, discounts, shipping, subscriptions, and repeat purchasing.

The project transforms raw customer shopping data into clear analytical results and an interactive dashboard that can support data-driven business decisions.

GitHub Repository

<https://github.com/Adithyasajeev22/Customer-Shopping-Behaviour-Analysis>